

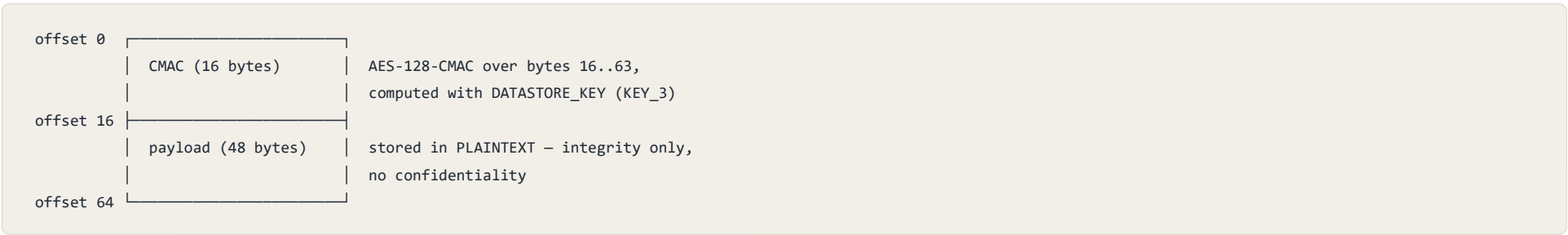
DATASTORE_KEY & Data-at-Rest Integrity

Compressed essence of Lesson 0002. Companion to the Key Slot Map.

1 • Identity card

ATTRIBUTE	VALUE	ANCHOR
Slot / raw KEY_ID	KEY_3 / 0x06	crypto_library_type.h:226 · Cry_30_Rh850Icus_Types.h:73
Origin of value	ICUSE TRNG, generated on-target, never injected or derived	secure_boot.c:433 → crypto_library.c:182
Registration	SHE M1 M2 M3 via CMD_LOAD_KEY, self-authenticated (AuthID = ID, empty slot ⇒ KEY_AuthID ≡ 0 ¹²⁸)	secure_boot.c:437–455 · manual §4.4 p. 43
Counter	1 (first registration, monotonic thereafter)	secure_boot.c:441
Flags	KU WP — CMAC-only + permanently frozen	secure_boot.c:442 · manual Table 4.2
Allowed operations	SecLib_MACGenerate / SecLib_MACVerify only (CMAC-AES-128)	crypto_library.c:243, 324
Created when	Step 2 of SecSB_EnableSecurity , after secure boot enable	secure_boot.c:145

2 • Secure block layout (64 bytes)



BLOCK_SIZE =64 (crypto_library.h:49), MAC_BYTE_SIZE =16. Flows: data_protection.c:48 (verify), :120 (generate).

3 • Flows

PATH	WRITE	READ
hsto_WriteSecure / hsto_ReadSecure (direct API, data_protection.c)	CMAC(payload) → header; write master block; on flash fail → backup at target + BLOCK_SIZE	read master → verify CMAC; fail & secure boot off → retry backup; fail & secure boot on → zeroize + error (lines 79–86)
NvM callbacks (nvmm_driver.c)	SecLib_NVM_MACVerify_Write — CMAC embedded before NvM write (line 541)	SecLib_NVM_MACVerify_Read PostReadTransformation — verify; fail ⇒ NvM reports INTEGRITY_FAILED (line 596)

Protected NvM blocks: SAM (MAC | data), Dataset (security log, two CMACs), ODOM_01–10 (CLI variant). One key seals them all.

4 • The two cryptographic formulas

Seal / verify (every day use)

CMAC = AES-128-CMAC(DATASTORE_KEY, payload[0..47]) → stored in block[0..15]

Registration transport (once, at provisioning)

K1 = KDF(KEY_AuthID, KEY_UPDATE_ENC_C) K2 = KDF(KEY_AuthID, KEY_UPDATE_MAC_C)

M2 = ENC_CBC,K1,IV=0 (Counter(28) | KEY_FLAG(6) | "0..."94 | KEY(128))

M3 = CMAC_K2 (M1 | M2)

SHE KDF(K, C): OUT_1 = ENC_ECB(OUT_0, K) ⊕ K ⊕ OUT_0 (OUT_0 = 0¹²⁸)

out = ENC_ECB(OUT_1, C) ⊕ C ⊕ OUT_1

KEY_UPDATE_ENC_C = 01015348 45008000 00000000 000000B0h

KEY_UPDATE_MAC_C = 01025348 45008000 00000000 000000B0h

Implemented in key_manager.c:329 (SecLib_CalculateKdfSHE — runs on ICUSE by bouncing derived keys through the RAM_KEY slot), :182 (SecLib_GenerateM2), :273 (SecLib_GenerateM3). Matches HIS SHE spec / manual §4.4.

5 • M2 byte map (what the unwrapping sees)

BYTE	0	1	2	3	4	5–15	16–31
Content	counter<<4 flags[7:4] (BE) → 00 00 00 19 for counter=1, KU WP (0x10 0x09)				flags[3:0]<<4	00...00	new key (TRNG)

key_manager.c:206–230. After CBC encryption with K1, IV=0. ICUSE inverts this inside the secure unit.

6 • Security ledger

PROPERTY	STATUS	NOTE
Integrity	✔	bit flips anywhere ⇒ verify fails ⇒ zeroize / INTEGRITY_FAILED
Authenticity	✔	forgery needs the slot key; it can't leave ICUSE, is CMAC-only (KU), frozen (WP)
Confidentiality	✖	payload plaintext by design; pair with an encryption key if needed
Freshness	✖	whole-block replay of older valid copies is undetected; odometer blocks address rollback at the application layer

7 • Variant & debugging notes

- KEYRAM_AS_KEY : DATASTORE_KEY ≡ RAM_KEY (volatile). data_protection.c:7 hard-codes a dev-only key (pKeyRam[]) reloaded on every read/write — **never for production**; the default build's TRNG + WP flow is the secure one.
- Slot empty at verify time ⇒ ICUS ERC_KEY_EMPTY (0x008) → E_VC_SECB00T_ICUS_KEY_EMPTY .
- Re-provision attempt after WP ⇒ ERC_KEY_WRITE_PROTECTED (ERR5) — enable path treats this as "already enabled" (secure_boot.c:142).
- M1M2M3 rejected (CMAC/counter/UID) ⇒ ERC_KEY_UPDATE_ERROR (ERR6).